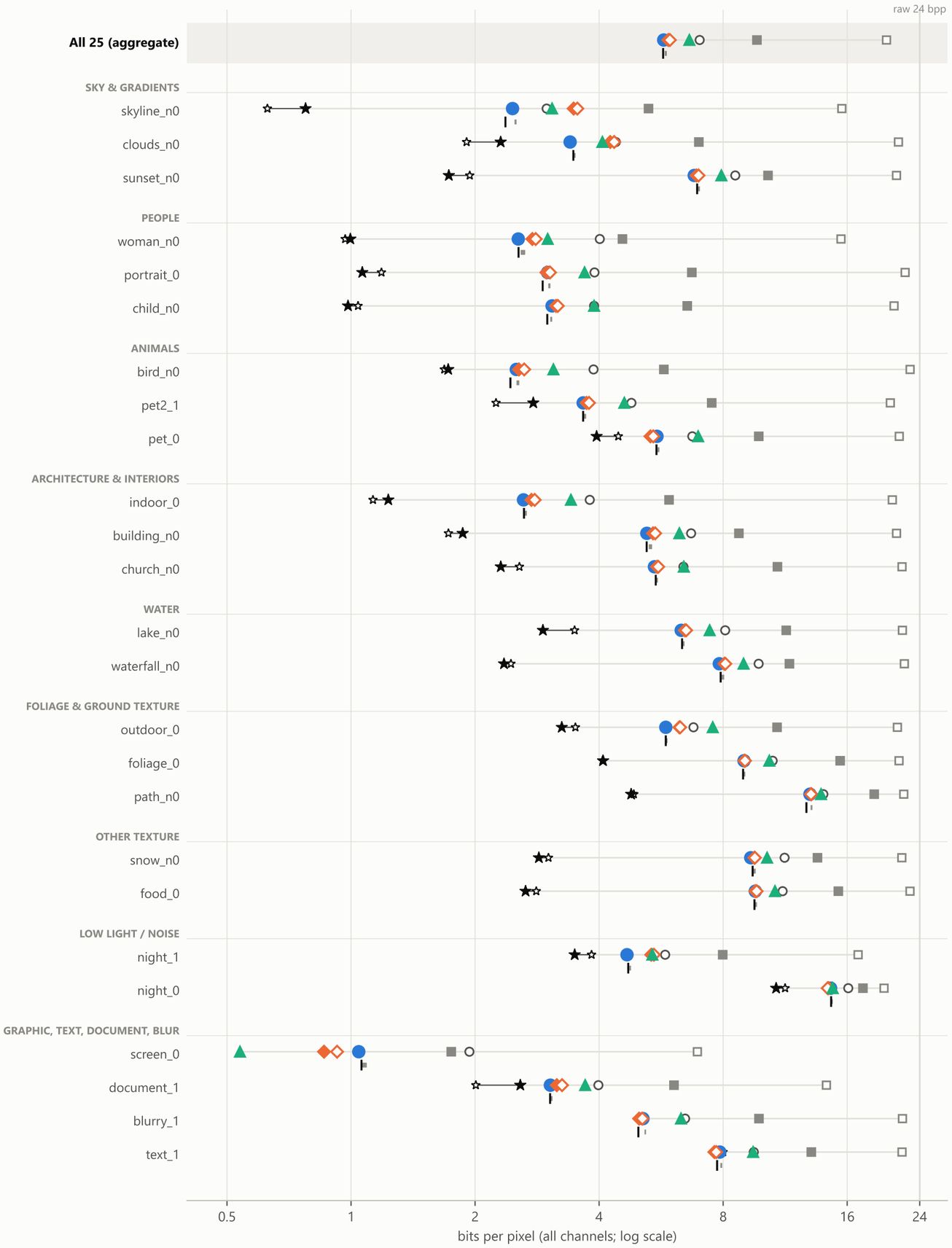
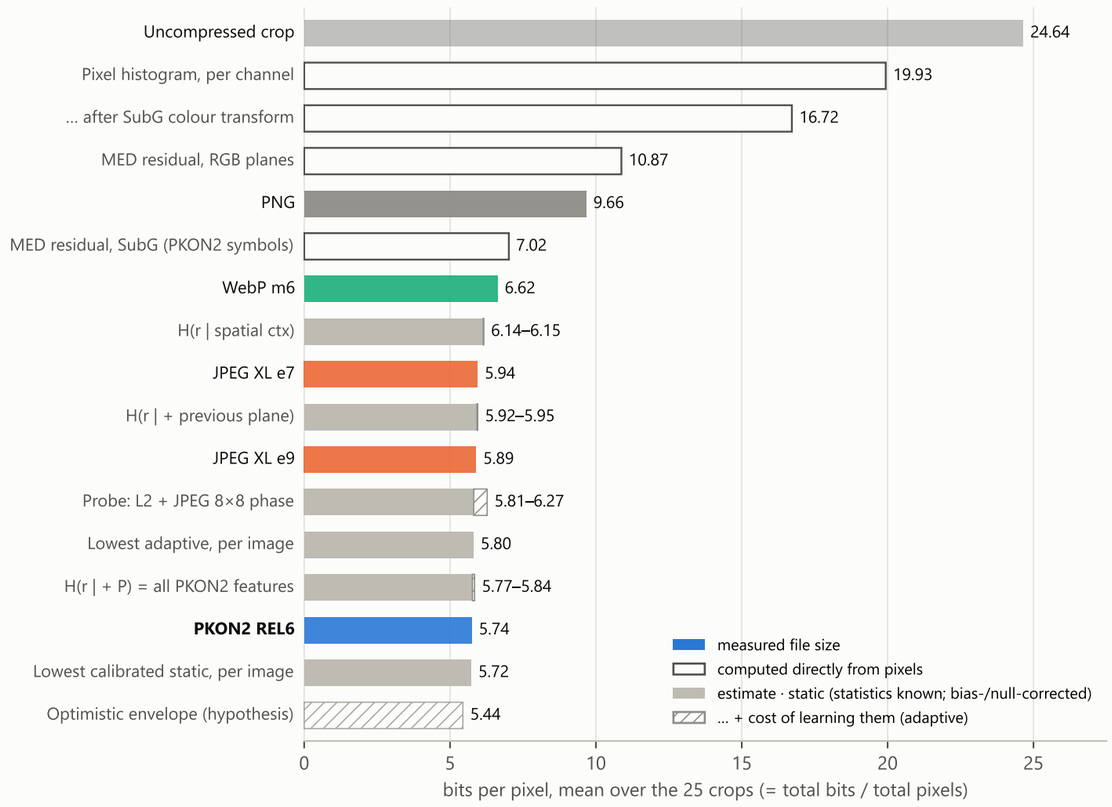
PKON2 compression landscape — reference map (bench25 diversity set, 25 × 1024² crops)

Measured: results/20261028-015857Z-bench25-diversity. Estimates: this directory's data/, computed from crops verified byte-identical (SHA-256) to the benchmark inputs.

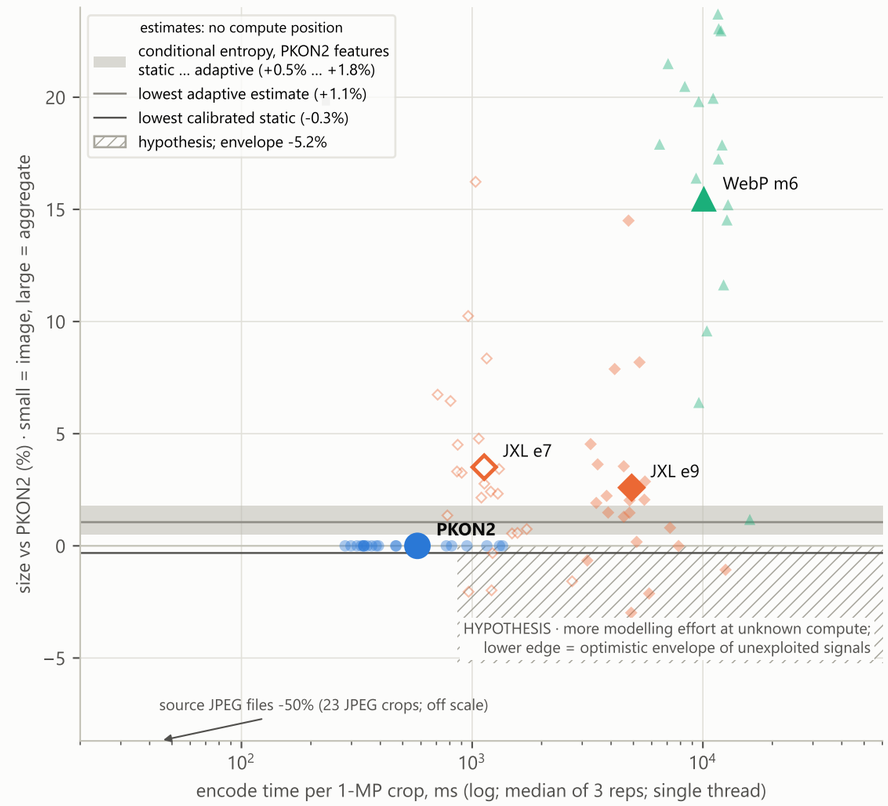
A + B · Per image: measured codecs (colour) against entropy references (grey/black)



B · Aggregate information ladder: what each description needs



C · Practical frontier: measured points, estimate band, hypothesis region



MEASURED (bench25, 1024×1024 centre crops)

- PKON2 REL6 (codec 3)
- JPEG XL lossless e9
- JPEG XL lossless e7
- WebP lossless m6
- PNG (png-rs best)

COMPUTED / ESTIMATED FROM THE SAME CROPS

- Zero-order pixel histogram, ΣH(channel) [computed]
- Zero-order entropy of PKON2's MED/SubG residual [computed]
- Conditional entropy given all PKON2 causal features (track under the codecs): static, null-calibrated (left) ... adaptive (right) [estimate]
- Lowest calibrated static estimate, any single probe [estimate]
- Source JPEG file, bits per decoded pixel (whole photo) [measured file size]
- ... rescaled to the crop's relative complexity [rough estimate]

CONCEPTUAL (NOT MEASURED)

- Hypothesis region / optimistic envelope — no measured points inside
- Estimate band drawn without a compute position

No value on this map is a lower bound on achievable size. Entropy estimates are relative to their model class; a richer model can go below them.
'Static' = statistics known for free (bias- and null-corrected);
'adaptive' = learned from the image itself (Dirichlet/KT code length).